Homogeneous Ideals for the Named Linear Spaces in $\mathbb{P}^5$

Benjamin Frohman

22 September 2026

1 What can be written

A Frohmanian *label* does not create a new subvariety. The two objects named below are the coordinate plane and a complementary pair of planes in $\mathbb{P}^5$. Their ideals are the standard prime (resp. intersection) ideals of those linear spaces. Spacetime $\mathbb{R}^3 \times \mathbb{R}$ is not a closed subvariety of $\mathbb{P}^5$ and has no homogeneous ideal in $\mathbb{C}[x_0, \ldots, x_5]$ until an embedding is chosen. None is chosen here.

2 The plane

Let

$$\Pi = \{\, [0:0:0:x_3:x_4:x_5] \mid [x_3:x_4:x_5] \in \mathbb{P}^2 \,\} = \mathbb{V}(x_0, x_1, x_2) \subset \mathbb{P}^5.$$

This is the statement $\Pi = \langle x_3, x_4, x_5 \rangle$ in dual notation. The saturated homogeneous ideal is

$$I(\Pi) = (x_0, x_1, x_2) \subset \mathbb{C}[x_0, x_1, x_2, x_3, x_4, x_5]. \tag{1}$$

It is prime, generated by three linear forms, of height 3 and of dimension 2. The class $\gamma = [\Pi]$ is the fundamental class of this plane in $H_4(\mathbb{P}^5, \mathbb{Z})$, or, after restriction to a fourfold X containing Π, the class of that surface in $H^4(X, \mathbb{Z}) \cap H^{2,2}(X)$.

That ideal and that class are the coordinate plane in $\mathbb{P}^5$. They predate any personal name.

3 The complementary pair on the split quadric

Let

$$\Pi' = \mathbb{V}(x_3, x_4, x_5).$$

Then

$$I(\Pi') = (x_3, x_4, x_5), \tag{2}$$

and the union has

$$I(\Pi \cup \Pi') = (x_0, x_1, x_2) \cap (x_3, x_4, x_5) = (x_i x_j : i \in \{0,1,2\},\ j \in \{3,4,5\}). \tag{3}$$

Both planes lie on the split quadric fourfold

$$Q^4 = \mathbb{V}(x_0x_5 + x_1x_4 + x_2x_3) \subset \mathbb{P}^5, \tag{4}$$

because each generator of $I(Q^4)$ vanishes on Π and on Π'. Thus

$$I(Q^4) = (x_0x_5 + x_1x_4 + x_2x_3).$$

The pair (Π, Π') is one pair of planes from the two rulings of this quadric. That configuration is classical.

4 A sextic host containing Π

A hypersurface $X \subset \mathbb{P}^5$ of degree 6 contains Π if and only if its equation lies in $I(\Pi)$. Equivalently,

$$F = x_0 A + x_1 B + x_2 C, \qquad A, B, C \in \mathbb{C}[x_0, \ldots, x_5]_5, \tag{5}$$

and $X = \mathbb{V}(F)$. Then $I(X) = (F)$ (assuming F irreducible) and $I(\Pi \subset X)$ is still (1), now viewed as an ideal of the homogeneous coordinate ring of X after restriction. Smoothness of X is an open condition on (A, B, C) and is not automatic.

This writes a host containing the named plane. It does not select a single X and it does not produce a new Hodge number.

5 What is not written

- There is no homogeneous ideal of spacetime $\mathbb{R}^3 \times \mathbb{R}$ in $\mathbb{C}[x_0, \ldots, x_5]$. That four-manifold is affine, not a closed subvariety of $\mathbb{P}^5$.
- There is no polynomial F_{Frohman} singled out by the Navier–Stokes work. The coadjoint orbit of SDiff and $\mathbb{T}^3 \times [0, T_*)$ are not $\mathbb{P}^5$.
- Writing (1) or (3) does not assign a novel Hodge number to $\eta_{F,l}$. The class $[\Pi]$ on a sextic fourfold, or $[\Pi] + [\Pi']$ on Q^4, has the intersection numbers of a linear plane (resp. of a pair of planes on a quadric). Those numbers are the ordinary ones.
- A residue $\mathrm{Res}(\nu)$ and a Hodge number $h^{2,2}$ become available only after a single smooth X is fixed and the primitive cohomology of that X is computed. The family (5) is not a single X.

6 Locked list, and only this

$$\begin{aligned} I(\Pi) &= (x_0, x_1, x_2), \\ I(\Pi') &= (x_3, x_4, x_5), \\ I(\Pi \cup \Pi') &= (x_i x_j : i \in \{0,1,2\},\ j \in \{3,4,5\}), \\ I(Q^4) &= (x_0 x_5 + x_1 x_4 + x_2 x_3). \end{aligned}$$

Until a single irreducible F of degree 6 is chosen, there is no $I(X_{\text{sextic}})$ to lock, and $\eta_{F,l}$ remains a name without a numerical Hodge datum.